Further Strategies — problem 14.44

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Let $f(x) = 4x - x^2$. For $x_0 \in \mathbb{R}$ we consider the infinite sequence $x_0, x_1 = f(x_0), x_2 = f(x_1) \ldots$. Prove that there exist infinitely many x_0, so that $x_0, x_1, x_2, \ldots$ consists of finitely many different values.